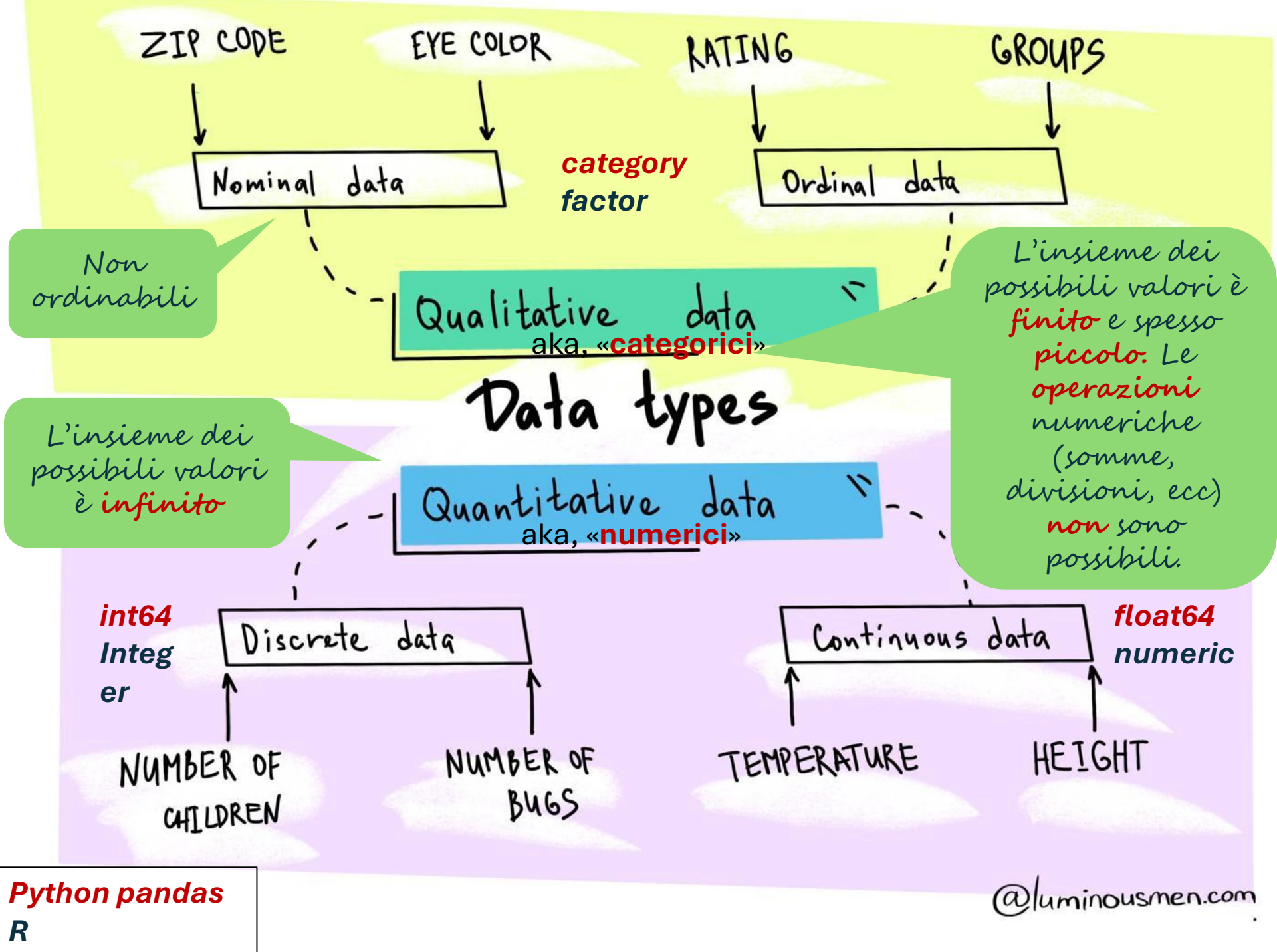
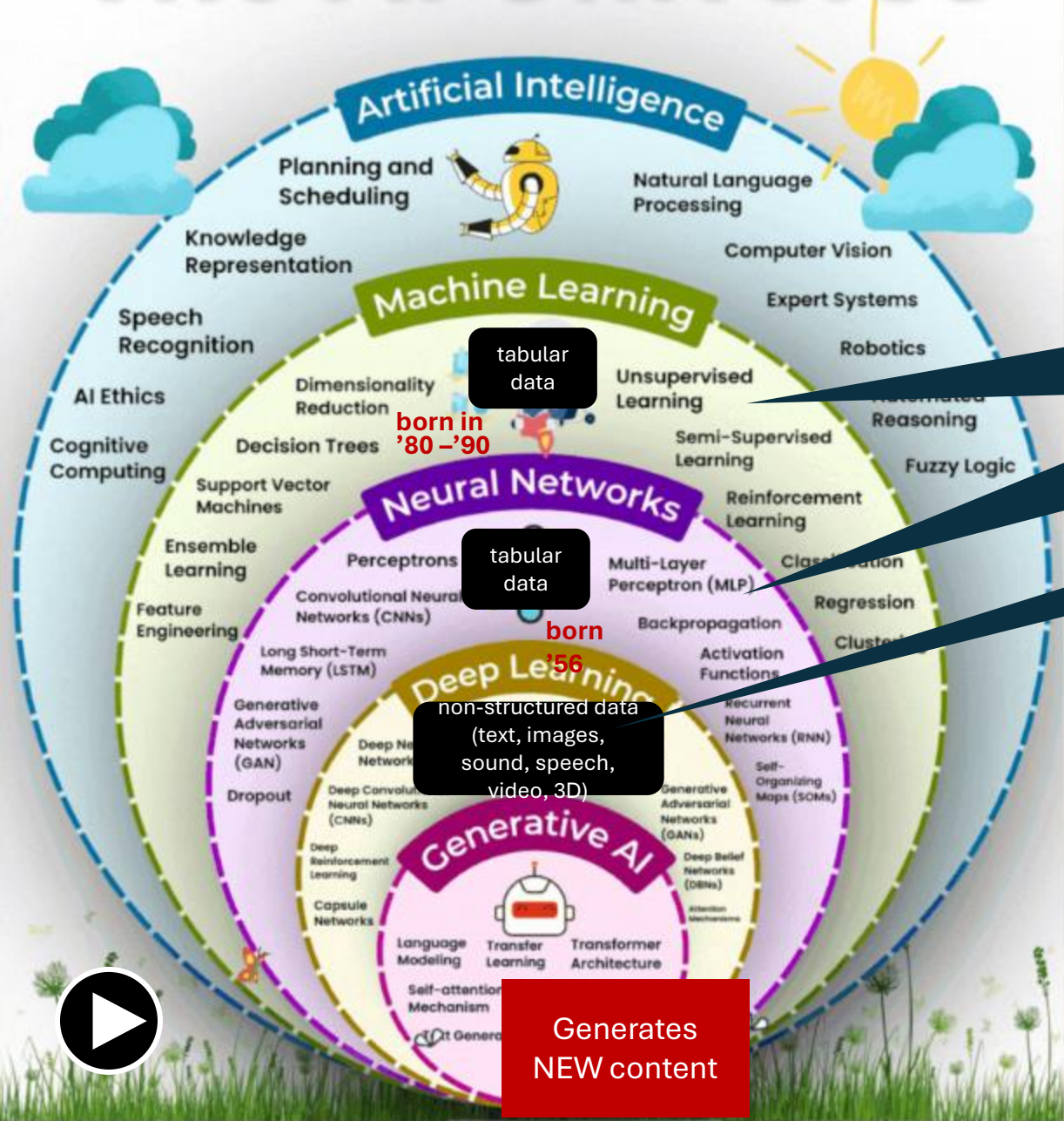




The AI Universe

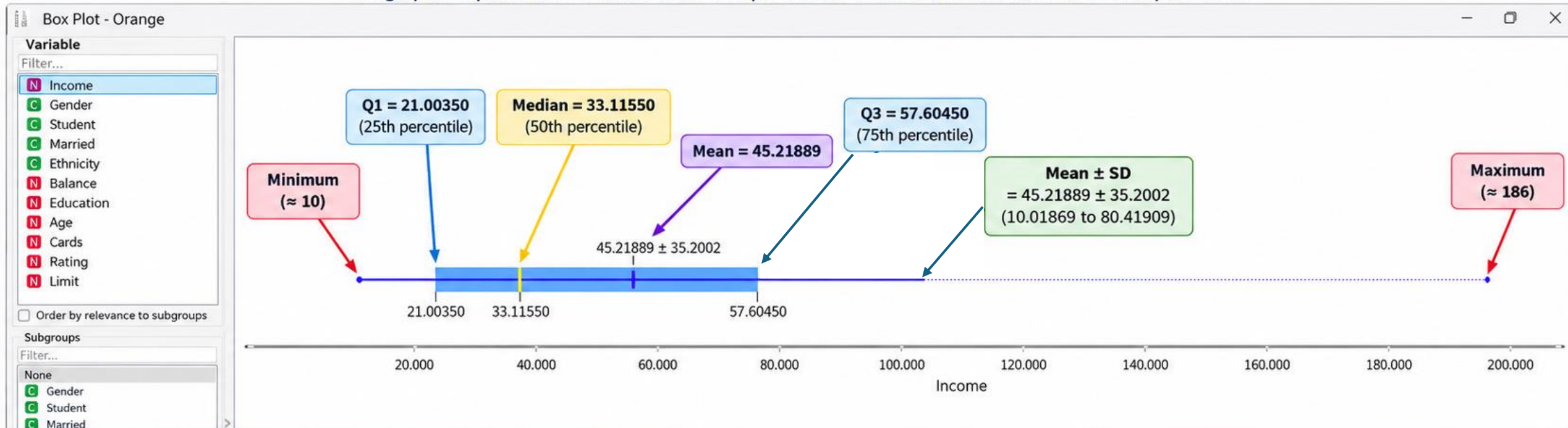


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Predict business phenomena i («predictive analyses»),
Operational Decisions

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Understanding the Orange Box Plot: Income

This graphic explains the elements of the box plot for the variable **Income** and how to interpret them.



Key Elements of the Box Plot

Minimum	≈ 10	Far-left end of the horizontal range (approximate).
Q1 (25th %)	21.00350	Left edge of the blue box.
Median	33.11550	Yellow vertical line inside the box.
Q3 (75th %)	57.60450	Right edge of the blue box.
Mean	45.21889	Dark blue vertical marker.
Mean ± SD	45.21889 ± 35.2002	Thin solid blue segment around the mean (range: 10.01869 to 80.41909).
Maximum	≈ 186	Far-right end of the horizontal range (approximate).

What the Plot Shows

	Blue box (IQR)	Represents the interquartile range (IQR), from Q1 to Q3, containing the middle 50% of the data.
	Yellow line	The median (50th percentile).
	Dark blue line	The mean (average).
	Thin solid blue segment	The mean ± 1 standard deviation (SD).
	Full horizontal range	Spans the minimum to maximum values (≈ 10 to ≈ 186).

Interpretation

- The distribution is **right-skewed**.
- The **mean (45.21889)** is **above the median (33.11550)**.
- The **right tail** is much longer than the left, suggesting a **high-end outlier** or extreme value in the data.
- Most incomes lie between 21.00350 and 57.60450 (the IQR), but there are some much higher values.

Important Note

Orange's box plot display is not the classical Tukey whisker convention. It shows the full range (minimum to maximum) with a box for the IQR and a separate mean $\pm$ SD segment.

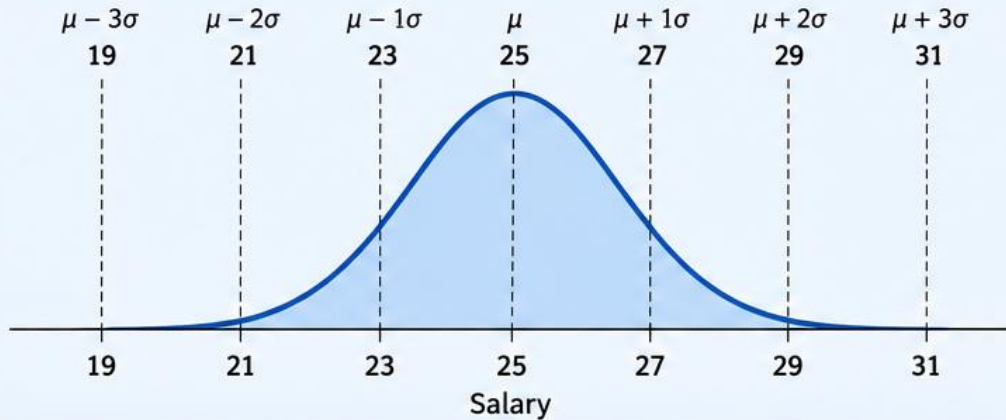
Two companies, same average salary, different variability

Both salary distributions are Normal (Gaussian) with mean $\mu = 25$.

Company A has $\sigma = 2$, Company B has $\sigma = 4$.

Company A

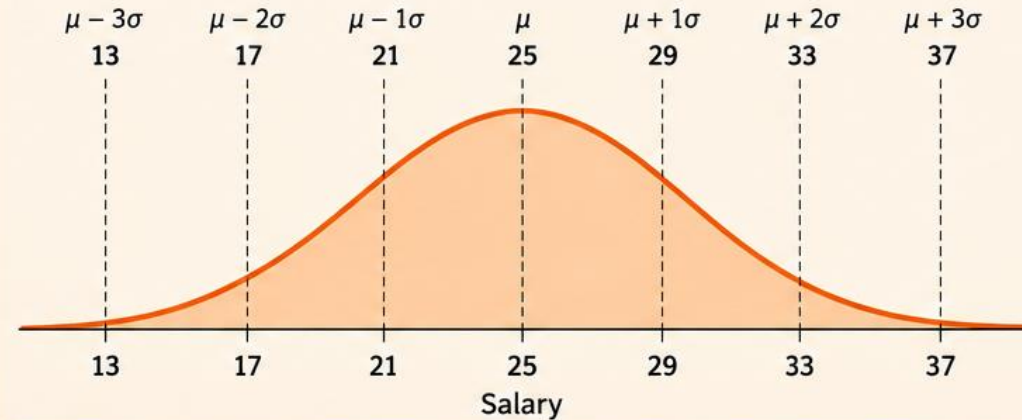
$\mu = 25$ | $\sigma = 2$



Range	Salary interval	Percentage of employees
$\mu \pm 1\sigma$	23 to 27	68.27%
$\mu \pm 2\sigma$	21 to 29	95.45%
$\mu \pm 3\sigma$	19 to 31	99.73%

Company B

$\mu = 25$ | $\sigma = 4$



Range	Salary interval	Percentage of employees
$\mu \pm 1\sigma$	21 to 29	68.27%
$\mu \pm 2\sigma$	17 to 33	95.45%
$\mu \pm 3\sigma$	13 to 37	99.73%

Key takeaway

Both companies have the same average salary ($\mu = 25$), so the percentage of employees within $\pm 1\sigma$, $\pm 2\sigma$ and $\pm 3\sigma$ is the same (**68.27%**, **95.45%**, **99.73%**).

However, Company B ($\sigma = 4$) has a larger spread of salaries than Company A ($\sigma = 2$): salaries are more variable and cover a wider range.

Same average.
Different variability.

A larger standard deviation means salaries are more spread out around the mean.